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PORs testing in multiple power domain devices

Abstract

According to an embodiment, a method for testing multiple power-on-resets in a system-on-chip with a multi-power domain architecture operating under a dual power flow mode is provided. The method includes powering up the system-on-chip to full power mode, decoupling a third power domain from a first power domain and a second power domain, monitoring a general purpose input/output (GPIO) pad of the third power domain during a ramping down of a supply of the third power domain, and detecting a logic transition at the GPIO pad of the third power domain corresponding to a trip-point of the power-on-reset of the third power domain.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to an electronic device and, in particular embodiments, to power-on-reset (POR) testing in multiple power domain devices.

BACKGROUND

(2) Systems on a chip may include multiple power domains, such as a RUN mode, a low-power, and a standby (or always-on) power domain. The device may switch between the different power domains to achieve energy efficiency. The standby power domain has very low power consumption and remains ON after the device is powered up. This allows the device to enter standby mode until, for example, a user's input triggers an end to the standby mode. The low-power power domain allows the device to operate in low-power mode—the low-power power domain is OFF in the standby mode. This allows the device to perform certain functions at reduced power consumption. The RUN mode power domain is typically used for performing full-power operations when the maximum performance of the device is required. The RUN power domain is OFF in the low-power and standby modes.

(3) Power-on-reset (POR) is commonly used to automatically reset a device or a region of a device. Power-on-reset is the first reset source, which gates all other reset sources within the device or at a particular region. Power-on-reset allows the device to start in a known, stable state and prevents it from operating in an unknown or undefined state.

(4) Testing a power-on-reset trip point (i.e., threshold check) is mandatory to ensure correct device functionality. In an electronic device with a multi-power domain operating in a dual power flow mode, each power domain includes an associated power-on-reset. This is in contrast to the conventional electronic device with a multi-power domain operating in a single-power follow mode or an electronic device with a single-power domain, where the electronic device operates with a single power-on-reset only, for example, in the standby power domain. A method, circuit, and system to test the various power-on-resets in an electronic device with a multi-power domain operating in a dual power flow are, thus, desirable.

SUMMARY

(5) Technical advantages are generally achieved by embodiments of this disclosure which describe the power-on-reset testing in multiple power domain devices.

(6) A first aspect relates to a method for testing multiple power-on-resets in a system-on-chip with a multi-power domain architecture operating under a dual power flow mode. The method includes powering up the system-on-chip to full power mode, decoupling a third power domain from a first power domain and a second power domain, monitoring a general purpose input/output (GPIO) pad of the third power domain during a ramping down of a supply of the third power domain, and detecting a logic transition at the GPIO pad of the third power domain corresponding to a trip-point of the power-on-reset of the third power domain.

(7) In a first implementation form of the method according to the first aspect as such, the method further includes decoupling the second power domain from the first power domain, monitoring a GPIO pad of the first power domain during a ramping down of a supply of the second power domain, and detecting a logic transition at the GPIO pad of the first power domain corresponding to a power-on-reset trip-point of the second power domain.

(8) In a second implementation form of the method according to the first aspect as such or any preceding implementation form of the first aspect, the method further includes monitoring a second GPIO pad of the first power domain during a ramping down of a supply of the first power domain

and detecting a logic transition at the second GPIO pad of the first power domain corresponding to a trip-point of the power-on-reset of the first power domain.

(9) In a third implementation form of the method according to the first aspect as such or any preceding implementation form of the first aspect, a supply of the second power domain and a supply of the first power domain remains stable during the ramping down of the supply of the third power domain.

(10) In a fourth implementation form of the method according to the first aspect as such or any preceding implementation form of the first aspect, the system-on-chip with the multi-power domain architecture operating under dual power flow mode has a first power flow direction during a power up or a reset sequence different from a second power flow direction during an exiting of a standby mode.

(11) In a fifth implementation form of the method according to the first aspect as such or any preceding implementation form of the first aspect, the first power flow direction is a sequential powering up of the third power domain, the second power domain, and the first power domain.

(12) In a sixth implementation form of the method according to the first aspect as such or any preceding implementation form of the first aspect, the second power flow direction is a sequential powering up of the second power domain and the third power domain, wherein the first power domain remains on during the standby mode.

(13) In a seventh implementation form of the method according to the first aspect as such or any preceding implementation form of the first aspect, the third power domain is used to perform full-power operations of the system-on-chip, the second power domain is used to perform low-power operations of the system-on-chip, and the first power domain is used to enter the system-on-chip into a standby mode.

(14) A second aspect relates to a system-on-chip. The system-on-chip includes a first power domain for operating the system-on-chip in a standby mode, the first power domain comprising a first switch, a second power domain for operating the system-on-chip in a low-power mode, the first switch configured to couple the first power domain to the second power domain, the second power domain comprising a second switch, a third power domain for operating the system-on-chip in a full operation mode, the second switch configured to couple the third power domain to the second power domain and the first switch configured to couple the second power domain to the first power domain. The third power domain includes a first level shifter and a second level shifter. The first level shifter is configured to generate a control signal with a logic level high in response to a supply of the third power domain being greater than a first threshold, the system-on-chip being fully powered up, and the second switch being in an open position, and generate the control signal with a logic level low in response to the supply of the third power domain being less than the first threshold, the system-on-chip being powered up in the first power domain and the second power domain and unpowered in the third power domain, and the second switch being in the open position. The second level shifter is configured to receive the control signal, generate an output signal with a logic level low in response to the control signal having a logic level high, and generate an output signal with a logic level high in response to the control signal having a logic level low.

(15) In a first implementation form of the system-on-chip according to the second aspect as such, the system-on-chip includes a GPIO pad coupled to an output of the second level shifter. By monitoring a transition from the logic level low to a logic level high at the GPIO pad during a ramping down of the supply of the third power domain, a power-on-reset trip point of the supply of the third power domain is detected.

(16) In a second implementation form of the system-on-chip according to the second aspect as such or any preceding implementation form of the second aspect, the second power domain includes a third level shifter configured to generate an output signal with a logic level high in response to a supply of the second power domain being greater than a second threshold, and generate an output

signal with a logic level low in response to a supply of the second power domain being less than the second threshold.
(17) In a third implementation form of the system-on-chip according to the second aspect as such or any preceding implementation form of the second aspect, the system-on-chip further includes a GPIO pad coupled to an output of the third level shifter, wherein by monitoring a transition from the logic level high to a logic level low at the GPIO pad during a ramping down of the supply of the second power domain, a power-on-reset trip point of the supply of the second power domain is detected.
(18) In a fourth implementation form of the system-on-chip according to the second aspect as such or any preceding implementation form of the second aspect, the first power domain includes a fourth level shifter and a fifth level shifter. The fourth level shifter is configured to generate a second control signal with a logic level high in response to a power-on-reset of the first power domain being greater than a third threshold, and generate the second control signal with a logic level low in response to the power-on-reset of the first power domain being less than the third threshold. The fifth level shifter is configured to receive the second control signal, generate an output signal with a logic level low in response to the second control signal having a logic level high, and generate an output signal with a logic level high in response to the second control signal having a logic level low.
(19) In a fifth implementation form of the system-on-chip according to the second aspect as such or any preceding implementation form of the second aspect, the system-on-chip further includes a GPIO pad coupled to an output of the fifth level shifter. By monitoring a transition from the logic level low to a logic level high at the GPIO pad during a ramping down of the supply of the first power domain, a power-on-reset trip point of the supply of the first power domain is detected.
(20) A third aspect relates to a method for testing multiple power-on-resets in a system-on-chip with a multi-power domain architecture operating under a dual power flow mode. The method includes having a first power domain for operating the system-on-chip in a standby mode, the first power domain comprising a first switch, having a second power domain for operating the system-on-chip in a low-power mode, the first power domain couplable to the second power domain using the first switch, and the second power domain comprising a second switch, having a third power domain for operating the system-on-chip in a full operation mode, the third power domain couplable to the second power domain using the second switch and further couplable to the first power domain using the first switch, operating the system-on-chip in a power-on-reset test mode, opening the second switch to decouple the third power domain from the first power domain and the second power domain during full power up of the system-on-chip, loading a test data register with a bit having logic level high, ramping down of a supply of the third power domain, and detecting a logic level transition at an output of a first level shifter, the first level shifter coupled to an inverted value of the test data register and controlled by a control signal generated by a second level shifter, the second level shifter generating a first control signal in response to the supply of the third power domain being greater than a threshold and generating a second control signal in response to the supply of the third power domain being less than the threshold.
(21) In a first implementation form of the method according to the third aspect as such, the detecting the logic level transition at the output of the first level shifter corresponds to a power-on-reset trip point of the supply of the third power domain.
(22) In a second implementation form of the method according to the third aspect as such or any preceding implementation form of the third aspect, the method further includes opening the first switch to decouple the second power domain from the first power domain, ramping down of a supply of the second power domain, and detecting a logic level transition at an output of a third level shifter, the third level shifter generating a first output signal with a logic level high in response to the supply of the second power domain being greater than a second threshold and generating a second output signal with a logic level low in response to a supply of the second

power domain being less than the second threshold.
(23) In a third implementation form of the method according to the third aspect as such or any preceding implementation form of the third aspect, the detecting the logic level transition at the output of the third level shifter corresponds to a power-on-reset trip point of the supply of the second power domain.
(24) In a fourth implementation form of the method according to the third aspect as such or any preceding implementation form of the third aspect, the method further includes loading a second test data register with a bit having logic level high, ramping down of a supply of the first power domain, and detecting a logic level transition at an output of a fourth level shifter, the fourth level shifter coupled to an inverted value of the second test data register and controlled by a second control signal generated by a fifth level shifter, the fifth level shifter generating a third control signal in response to the supply of the first power domain being greater than a third threshold and generating a fourth control signal in response to the supply of the first power domain being less than the third threshold.
(25) In a fifth implementation form of the method according to the third aspect as such or any preceding implementation form of the third aspect, the detecting the logic level transition at the output of the fourth level shifter corresponds to a power-on-reset trip point of the supply of the first power domain.
(26) Embodiments can be implemented in hardware, software, or in any combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a more complete understanding of the present disclosure and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a block diagram of an embodiment system-on-chip (SoC);
(3) FIG. **2** is a block diagram of a circuit for testing the power-on-reset of a system-on-chip having a multi-power domain architecture that operates under a single power flow direction during either boot-up or exiting from standby mode;
(4) FIG. **3** is a block diagram of an embodiment power-on-reset circuit for testing the power-on-reset trip-point of the third power domain of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode;
(5) FIG. **4** is a block diagram of an embodiment power-on-reset circuit for testing the power-on-reset trip-point of the second power domain of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode;
(6) FIG. **5** is a block diagram of an embodiment power-on-reset circuit for testing the power-on-reset trip-point of the first power domain of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode;
(7) FIG. **6** is a block diagram of an embodiment circuit for generating the isolation control for the signals crossing from an unpowered domain to the power domain;
(8) FIG. **7** is a block diagram of an embodiment circuit used to illustrate how the input value to the third input of the second level shifter is forced to the value of “0”;
(9) FIG. **8** is a block diagram of an embodiment circuit used to illustrate the input value to the third input of the second level shifter;
(10) FIG. **9** is a flow chart of an embodiment method to test power-on-resets of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode;
(11) FIG. **10** is a block diagram of an embodiment processing system; and
(12) FIG. **11** is a block diagram of an embodiment circuit for testing power-on-resets of a system-

on-chip with a multi-power domain architecture operating under a dual power flow mode.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(13) This disclosure provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The particular embodiments are merely illustrative of specific configurations and do not limit the scope of the claimed embodiments. Features from different embodiments may be combined to form further embodiments unless noted otherwise.

(14) Variations or modifications described in one of the embodiments may also apply to others. Further, various changes, substitutions, and alterations can be made herein without departing from the spirit and scope of this disclosure as defined by the appended claims.

(15) While the inventive aspects are described primarily in the context of an electronic device with a multiple power domain of a quantity of three power domains, it should also be appreciated that these inventive aspects may also apply to an electronic device with a multiple power domain of any quantity of power domains.

(16) In embodiments, a method, circuit, and system are provided for testing multiple power-on-resets in a system-on-chip with a multi-power domain architecture operating under a dual power flow mode.

(17) Aspects of the disclosure include a system-on-chip that includes a first power domain, a second power domain, and a third power domain. The first power domain is used by the system-on-chip for operating in a standby mode. The second power domain is used by the system-on-chip for operating in a low-power mode. The third power domain is used by the system-on-chip for operating in a full operation mode.

(18) The first power domain includes a first switch. The second power domain includes a second switch. The first switch is configured to couple the first power domain to the second power domain and the second power domain to the first power domain. The second switch is configured to couple the third power domain to the second power domain. The third power domain includes a first level shifter and a second level shifter.

(19) The first level shifter is configured to generate a control signal with a logic level high in response to (i) a supply of the third power domain being greater than a first threshold, (ii) the system-on-chip being fully powered up, (iii) and the second switch being in an open position.

(20) The first level shifter is further configured to generate the control signal with a logic level low in response to (i) the supply of the third power domain being less than the first threshold, (ii) the system-on-chip being powered up in the first power domain and the second power domain and unpowered in the third power domain, and (iii) the second switch being in the open position.

(21) The second level shifter is configured to receive the control signal and generate an output signal with a logic level low in response to the control signal having a logic level high. The second level shifter is further configured to generate an output signal with a logic level high in response to the control signal having a logic level low.

(22) In embodiments, the system-on-chip includes a GPIO pad coupled to an output of the second level shifter. By monitoring a transition from the logic level low to a logic level high at the GPIO pad during a ramping down of the supply of the third power domain, a power-on-reset trip point of the supply of the third power domain is detected.

(23) In another embodiment, the second power domain includes a third level shifter configured to generate an output signal with a logic level high in response to a supply of the second power domain being greater than a second threshold, and generate an output signal with a logic level low in response to a supply of the second power domain being less than the second threshold.

(24) In yet another embodiment, the system-on-chip further includes a GPIO pad coupled to an output of the third level shifter. By monitoring a transition from the logic level high to a logic level low at the GPIO pad during a ramping down of the supply of the second power domain, a power-on-reset trip point of the supply of the second power domain is detected.

(25) In yet another embodiment, the first power domain includes a fourth level shifter and a fifth

level shifter. The fourth level shifter is configured to generate a second control signal with a logic level high in response to a power-on-reset of the first power domain being greater than a third threshold, and generate the second control signal with a logic level low in response to the power-on-reset of the first power domain being less than the third threshold. The fifth level shifter is configured to receive the second control signal, generate an output signal with a logic level low in response to the second control signal having a logic level high, and generate an output signal with a logic level high in response to the second control signal having a logic level low.

(26) In yet another embodiment, the system-on-chip further includes a GPIO pad coupled to an output of the fifth level shifter. By monitoring a transition from the logic level low to a logic level high at the GPIO pad during a ramping down of the supply of the first power domain, a power-on-reset trip point of the supply of the first power domain is detected. These and other details are further detailed below.

(27) FIG. **1** illustrates a block diagram of an embodiment system-on-chip (SoC) **100** as disclosed in U.S. Pat. No. 11,550,348, which is hereby incorporated by reference herein in its entirety. System-on-chip **100** is configured to have multiple power domains. System-on-chip **100** may be a component of an electronic device, such as a computer, a mobile phone, a home appliance, or the like. System-on-chip **100** includes a first power domain (PD0) **102**, a second power domain (PD1) **104**, and a third power domain (PD2) **106**, which may (or may not) be arranged as shown.

(28) Optionally, System-on-chip **100** may include a controller **108** operating in, for example, the third power domain **106**—to minimize taxing resources of the first power domain **102** and second power domain **104**.

(29) Also shown are RUN mode switch **120**, **122**, and **124**, and a low-power switch **118**. RUN mode switches **120**, **122**, and **124** are arranged between the third power domain **106** and the second power domain **104**. The low-power switch **118** is arranged between the first power domain **102** and the second power domain **104**. It should be appreciated that when the RUN mode switches **120**, **122**, and **124**, and low-power switch **118** are closed, all three low-voltage power domains effectively behave as a single low-voltage power domain.

(30) System-on-chip **100 100** may include a ring pad **125** where the I/O pads of the system-on-chip **100** are located. Power supply pads **126** on the ring pad **125** correspond to supply pads for the third power domain **106**. System-on-chip **100** may include additional components not shown in FIG. **1**. Further, system-on-chip **100** may have other power domains or supplies (not shown). The number of I/O pads illustrated in FIG. **1** for system-on-chip **100** is non-limiting, and embodiments with fewer or greater pads are similarly contemplated.

(31) First power domain **102**, second power domain **104**, and third power domain, may collectively be referred to as the power domains of the system-on-chip **100**. Each power domain may include its respective electrical components or circuits that perform certain functions. Each of the first power domain **102**, second power domain **104**, and third power domain **106** may operate on a low-voltage (LV) supply. First power domain **102** also includes a high-voltage (HV) supply to provide the various low-voltage supplies in the different power domains in one configuration of the system-on-chip **100**.

(32) The first power domain **102** can also be referred to as a standby power domain, an ultra-low-power (ULP) power domain, or an always-on power domain. The first power domain **102** is powered on during a standby mode of system-on-chip **100**. In embodiments, the first power domain **102** monitors user input during the standby mode, and in response to, for example, detecting a user input, exits the standby mode and enters into low power mode or RUN mode.

(33) The second power domain **104** can also be referred to as a low-power (LP) power domain. The second power domain **104** may be powered off during the standby mode of system-on-chip **100** when the first power domain **102** is powered on. In embodiments, the second power domain **104** may perform certain functions at reduced power consumption in the low-power (LP) power mode. In the low-power power mode, the first power domain **102** and the second power domain **104** are

powered on, and the third power domain **106** is powered off.
(34) The third power domain **106** can also be referred to as a RUN mode power domain or a switchable power domain. Third power domain **106** may be powered off during the standby mode of system-on-chip **100** when the first power domain **102** is powered on and the second power domain **104** is powered off. Third power domain **106** may be switchably powered on or powered off at other times. In embodiments, the third power domain **106** may be used for performing a full-power operation in the RUN mode when the maximum performance of the system-on-chip **100** is required. System-on-chip **100 100** may exit the low-power mode to enter the RUN mode. In the RUN mode, the first power domain **102**, the second power domain **104**, and the third power domain **106** are all powered on.
(35) In embodiments, a power domain is said to be powered on, or being powered, when a supply voltage is provided to the power domain to turn on (e.g., providing the supply voltage) circuits operating within the power domain. When a power domain no longer receives the supply voltage, the power domain is said to be powered off, turned off, or not powered.
(36) In embodiments, the first power domain **102** includes a power management unit (PMU) **110**. The power management unit **110** may include a switched-mode power supply (SMPS) **112** (e.g., buck converter, boost converter, etc.), a first regulator **114**, and a second regulator **116**.
(37) Switched-mode power supply **112** is switched on and off alternately by a respective control signal (e.g., a pulse-width modulated (PWM) control signal) to provide a supply voltage. Switched-mode power supply **112** may supply power to the first power domain **102**, the second power domain **104**, and the third power domain **106**. In embodiments, the first regulator **114** is powered off in response to the switched-mode power supply **112** powering the first power domain **102** and the second power domain **104**.
(38) In embodiments, the first regulator **114** is a linear voltage regulator configured to provide a low voltage supply. The first regulator **114** may have a ballast **132** located in the ring pad **125** of system-on-chip **100**—ballast **132** is internal to the ring pad **125**. In embodiments, ballast **132** includes a low-voltage (LV) supply pad **130** and a pad **134**. Low-voltage (LV) supply pad **130** may be coupled with a low-voltage supply package pin. Pad **134** may be coupled with a high-voltage supply package pin.
(39) In embodiments, the second regulator **116** is a linear regulator configured to provide an ultra-low voltage supply. The second regulator **116** may power the first power domain **102** during a standby mode of system-on-chip **100**. The second regulator **116** may have its loop closed within the power management unit **110**. A ballast for the second regulator **116** may be located within the power management unit **110**. In embodiments, the second regulator **116** is powered on only during the standby mode of system-on-chip **100**.
(40) In embodiments, the first power domain **102** receives power from the low-voltage (LV) supply pad **130**. In embodiments, the third power domain **106** receives power from the power supply pads **126**. In embodiments, the second power domain **104** receives power through the first power domain **102**, the third power domain **106**, or both the first power domain **102** and the third power domain **106** depending on the configuration of the low-power switch **118** and RUN mode switches **120**, **122**, and **124**. The low-power switch **118** and the RUN mode switches **120**, **122**, and **124** are controlled through respective pads within the first power domain **102**, which is always on once the device is powered up (i.e., allowing the switches to open and close regardless of whether the second power domain **104** or third power domain **106** are powered up).
(41) The second power domain **104** is decoupled from the first power domain **102** and the third power domain **106** in response to the low-power switch **118** and RUN mode switches **120**, **122**, and **124** being in an open position. In this configuration, the second power domain **104** is powered off.
(42) In response to the low-power switch **118** being closed and RUN mode switches **120**, **122**, and **124** being open, the first power domain **102** is coupled to the second power domain **104**, and the first regulator **114** may be powering both the first power domain **102** and the second power domain

104. It should be appreciated that additional switches may be used to couple the first power domain **102** and the second power domain **104**.
(43) In response to the low-power switch **118** being open and RUN mode switches **120**, **122**, and **124** being in a closed position, the third power domain **106** is coupled to the second power domain **104**, such that the switched-mode power supply **112** provides a supply voltage to both power domains. In embodiments, RUN mode switches **120**, **122**, and **124** are in a closed position in response to the switched-mode power supply **112** is powered. Although system-on-chip **100** is shown to have three RUN mode switches **120**, **122**, and **124**, the number of RUN mode switches are non-limiting, and fewer or greater RUN mode switches may be included in other embodiments.
(44) Finally, in response to the low-power switch **118** and the RUN mode switches **120**, **122**, and **124** being in a closed position, the second power domain **104** is coupled to the first power domain **102** and the third power domain **106**. In this configuration, switched-mode power supply **112** provides a supply voltage to the third power domain **106**, the second power domain **104** (through the RUN mode switches **120**, **122**, and **124**), and the first power domain **102** (through the low-power switch **118**).
(45) In embodiments, controller **108** is used to control the operation (open or closed position) of the low-power switch **118** and the RUN mode switches **120**, **122**, and **124**. In embodiments, the low-power switch **118** and the RUN mode switches **120**, **122**, and **124** are configured to be closed in response to no high voltage (HV) supply being made available to the switched-mode power supply **112**. In embodiments, controller **108** controls the operation of the first regulator **114** and the second regulator **116**.
(46) In embodiments, the switched-mode power supply **112** of the system-on-chip **100** is coupled to an external (e.g., off-chip component) circuit **142** through pad **140**. Circuit **142** may include, for example, a ballast, an inductor, and a capacitor. In response to the system-on-chip **100** being coupled to circuit **142**, the switched-mode power supply **112** provides a voltage supply at a package pin coupled to power supply pad **129**, which is coupled to power supply pads **126** through a power bar (between the package and the die of system-on-chip **100**). Thus, the switched-mode power supply **112** can (i) power the third power domain **106** and (ii) power the second power domain **104** and the first power domain **102** (depending on the configuration of the low-power switch **118** and the RUN mode switches **120**, **122**, and **124**).
(47) FIG. **2** illustrates a block diagram of a circuit **200** for testing the power-on-reset of a system-on-chip having a multi-power domain architecture that operates under a single power flow direction during either boot-up or exiting from standby mode. Conventionally, the power flow direction during boot-up is the same as the power flow direction during an exiting from standby mode.
(48) For example, for a system-on-chip having a multi-power domain architecture that operates under a single power flow direction, the boot-up sequence includes sequentially powering up the first power domain **102** (e.g., entering into standby mode), then the second power domain **104** (e.g., entering into low-power mode), and finally the third power domain **106** (e.g., entering into RUN mode). When the system-on-chip transitions from RUN mode to low-power mode, the third power domain **106** is powered off. Next, when the device transitions from the low-power mode to the standby mode, the second power domain **104** is powered off. The transition from standby mode to RUN mode includes first powering the second power domain (e.g., entering into low-power mode) and then powering the third power domain (e.g., entering into RUN mode) sequentially.
(49) As the conventional system-on-chip has a single power flow direction, the electronic device includes only a single power-on-reset **222**, which monitors the first power domain supply and controls (i.e., signal out) the power voltage status to the third power domain pads (e.g., GPIO pad **252** and other GPIO pads of the third power domain **106** and GPIO pad **228** and other GPIO pads of the first power domain **102**). In embodiments, the control signal (PWR_OK2) is coupled to all LV to HV level shifters before all GPIO pads of the third power domain **106**. In embodiments,

control signal (PWR_OK1) is coupled to all LV to HV level shifters before all GPIO pads of the first power domain **102**

(50) In embodiments, circuit **200** includes a first sub-circuit **220** and a second sub-circuit **240**. The first sub-circuit **220** is used to test the single power-on-reset **222** in the first power domain **102** and the second sub-circuit **240** is used to test the single power-on-reset **222** in the third power domain **106**.

(51) The first sub-circuit includes a single power-on-reset **222**, a first level shifter **224**, a second level shifter **226**, and a GPIO pad **228**. The second sub-circuit **240** includes a single power-on-reset **222**, a first switch **244**, a second switch **246**, a third level shifter **248**, a fourth level shifter **250**, and a GPIO pad **252**. Thus, single power-on-reset **222** can be checked at GPIO pad **228** within the first power domain **102**, and at GPIO pad **252** in the third power domain **106** when both first switch **244** and second switch **246** are in a closed position.

(52) In embodiments, the first switch **244** in circuit **200** is mapped to the low-power switch **118** in the system-on-chip **100**. In embodiments, the second switch **246** in circuit **200** is mapped to the RUN mode switches **120**, **122**, and **124** in the system-on-chip **100**. It is noted that, although multiple RUN mode switches are shown in FIG. **1**, the number of switches is reduced to one in FIG. **2** for simplicity of the discussion. Thus, the number of RUN mode switches is non-limiting, and fewer or greater than three switches are contemplated in embodiments. Further, although a single low-power switch **118** is shown in the various figures, the number of switches is non-limiting, and a greater number of switches may also be contemplated.

(53) The testing of the power-on-reset for a system-on-chip having a multi-power domain architecture that operates under a single power flow direction using circuit **200** includes (i) closing the first switch **244** and the second switch **246** and (ii) powering up the first power domain **102**, the second power domain **104**, and the third power domain **106**. This configuration effectively makes the system-on-chip operate as a single power domain device.

(54) In embodiments, circuits operating within, for example, each of the third and the first power domains include an IP that interfaces (e.g., Controller Area Network (CAN), Local Interconnect Network (LIN), etc.) with the external component through a respective general-purpose input/output (GPIO) pad after being level shifted by the level shifter coupled between the IP and the GPIO. The GPIO is only operable when activated by the control signal (PWR_OK). The control signal (PWR_OK) is only activated when the single power-on-reset in the first power domain **102** is asserted (i.e., above a threshold), regardless of the domain in which the IP is interfacing with the GPIO. It is noted that there are no GPIO pads in the second power domain **104**.

(55) The single power-on-reset **222** is coupled to the low-voltage (LV) supply of the system-on-chip **100**—the low-voltage (LV) supply is the common low-voltage supply common to all power domains when all power domains are powered up. Further, each level shifter **224**, **226**, **248**, and **250** has a first input (i.e., a first transistor of the level shifter) coupled to the low-voltage (LV) supply and a second input (i.e., a second transistor of the level shifter) coupled to a high-voltage (HV) supply.

(56) Regarding the first sub-circuit **220**, the low-voltage (LV) signal is received at a third input of the first level shifter **224** from the single power-on-reset **222**. In embodiments, the LV signal from single power-on-reset **222** is logic high when the low-voltage supply is above the single power-on-reset threshold. In embodiments, the LV signal from the single power-on-reset **222** is logic low when the low-voltage supply is below the single power-on-reset threshold. The low-voltage supply is provided from, for example, low-voltage (LV) supply pad **130**. The high-voltage supply is provided from, for example, pad **134**.

(57) The output of the first level shifter **224** is a level-shifted control signal (PWR_OK1) provided as a control signal to the second level shifter **226**. The control signal (PWR_OK1) has level shifted value as a function of the voltage level at the sense input of single power-on-reset **222**. In embodiments, the sense input of single power-on-reset **222** is the LV SUPPLY driven from low-

voltage (LV) supply pad **130**.

(58) In embodiments, the third input of the second level shifter **226** is coupled to an output of an IP of a circuit operating within the first power domain. The output of the second level shifter **226** is coupled to the GPIO pad **228** of the system-on-chip **100**. During the power-on-reset test, the value at the third input of the second level shifter **226** is forced to “0” by, for example, loading a test data register (TDR) bit through a JTAG—further detailed in FIG. **7**.

(59) Regarding second sub-circuit **240**, the low-voltage (LV) signal is received at a third input of the third level shifter **248** from the single power-on-reset **222**. The output of the third level shifter **248** is a control signal (PWR_OK2) for the fourth level shifter **250**. In embodiments, the third input of the fourth level shifter **250** is coupled to an output of an IP of a circuit operating within the third power domain. The output of the fourth level shifter **250** is coupled to the GPIO pad **252** of the system-on-chip **100**. During the power-on-reset test, the value at the third input of the fourth level shifter **250** is forced to “0”.

(60) Initially, to test the power-on-reset, the low-voltage (LV) supply at the single power-on-reset **222** is greater than or equal to a power-on-reset threshold. Thus, the first control signal (PWR_OK1) at the output of the first level shifter **224** and the second control signal (PWR_OK2) at the output of the third level shifter **248** are equal to “1”. In response to the first control signal (PWR_OK1) being activated (i.e., equal to “1”), the GPIO pad **228** is operable and equal to the value forced at the third input of the second level shifter **226** or “0”. Likewise, in response to the second control signal (PWR_OK2) being activated (i.e., equal to “1”), the GPIO pad **252** is operable and equal to the value forced at the third input of the fourth level shifter **250** or “0”.

(61) Next, the low-voltage (LV) is ramped down until the voltage value of the low-voltage (LV) supply is less than the power-on-reset threshold. As such, the first control signal (PWR_OK1) at the output of the first level shifter **224** and the second control signal (PWR_OK2) at the output of the third level shifter **248** transition from a value of “1” to “0”, and the GPIO pad **228** and the GPIO pad **252** are pulled up (due to the HV supply having a default value to satisfy certain application requirements—using, for example, a pull-up resistor tied to the HV supply). When the GPIO pads **228** and **252** are pulled up, the value at the pads transitions from “0” to “1”. Thus, the trip point for the single power-on-reset **222** is detected by monitoring the GPIO pads **228** and **252** and detecting the transition from “0” to “1”. In embodiments, a test device is coupled to the GPIO pads **228** and **252** to monitor the transition from “0” to “1” to test the power-on-reset of the circuit **200**.

(62) The reader's attention is directed to U.S. Pat. Nos. 9,698,771 and 9,941,875, which are incorporated herein by reference in their entirety, for a more detailed description of conventional circuits used to test the power-on-reset of a system-on-chip having a multi-power domain architecture that operates under a single power flow direction during either boot-up or exiting from standby mode.

(63) A detailed discussion of a system-on-chip having a multi-power domain architecture that operates under a dual power flow mode can be found in, for example, U.S. Pat. No. 11,513,544 B1, which is hereby incorporated by reference herein in its entirety.

(64) In the dual power flow mode, the power flow direction (i.e., first power flow path) in the system-on-chip **100** during power-up or a reset (e.g., boot-up sequence or power-up sequence) is different from the power flow direction (i.e., second power flow path) when the system-on-chip **100** exits the standby mode and enters the RUN mode.

(65) In a system-on-chip having a multi-power domain architecture that operates under dual power flow directions, the power-up or reset sequence includes sequentially powering up the third power domain **106**, then the second power domain **104**, and finally the first power domain **102** (i.e., first power flow path)—opposite to the single power flow direction illustrated in FIG. **2**.

(66) When the system-on-chip transitions from RUN to low-power mode, the third power domain **106** is powered off. Next, when the device transitions from the low-power mode to the standby

mode, the second power domain **104** is powered off. The transition from standby to RUN mode includes first powering the second power domain (e.g., entering into low-power mode) and then powering the third power domain (e.g., entering into RUN mode) in a sequential manner (i.e., the second power flow path)—similar to the single power flow direction illustrated in FIG. **2**.

(67) FIGS. **3-5** illustrate simplified block diagrams of embodiment power-on-reset circuits **300**, **400**, and **500** for testing the power-on-reset of the multiple-power domains in a system-on-chip operating under a dual power flow mode. In contrast to a system-on-chip with a single power-on-reset, in a system-on-chip with a multi-power domain architecture that operates under a dual power flow mode, each power domain includes a dedicated power-on-reset.

(68) The GPIO corresponding to a specific power domain is only operable when activated by a dedicated control signal (PWR_OK) that depends on the supply voltage level and the power-on-reset threshold of the associated power domain. Thus, the dedicated control signal is only activated when the supply voltage level exceeds the power-on-reset threshold for the associated power domain.

(69) In embodiments, the power-on-reset of the third power domain **106** is gated by the power-on-reset of the first power domain **102** and the second power domain **104** (i.e., it requires the power-on-reset of the first, second, and third power domains be greater than a threshold). Thus, a GPIO associated with the third power domain **106** is only operable when the power-on-resets of the first, second, and third power domains are asserted (i.e., each above the power-on-reset threshold). A GPIO associated with the first power domain **102** is operable when the power-on-reset of the first power domain **102** is asserted (i.e., above the power-on-reset threshold).

(70) Disadvantageously, circuit **200** cannot be used to test each dedicated power-on-reset. For example, when the first switch **244** and the second switch **246** are closed, each dedicated power-on-reset senses the same low voltage (LV) supply. Further, it would be unclear which power-on-reset causes a trip point when monitoring the GPIO pad **252** and detecting the transition from “0” to “1”. Moreover, monitoring the GPIO pad **228** can only determine the power-on-reset dedicated to the first power-on-domain. Embodiments of this disclosure provide a method, circuit, and system that cures these deficiencies in the prior art.

(71) FIG. **3** illustrates a block diagram of an embodiment power-on-reset circuit **300** for testing the power-on-reset trip-point of the third power domain **106** of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode. Power-on-reset circuit **300** includes a first power-on-reset **302**, a second power-on-reset **304**, a third power-on-reset **306**, a first switch **308**, a second switch **310**, an AND gate **312**, a first level shifter **314**, a second level shifter **316**, and a GPIO pad **318**, which may (or may not) be arranged as shown. Power-on-reset circuit **300** may include additional components not shown.

(72) In embodiments, the first power-on-reset **302** monitors the PD0_LV SUPPLY from low-voltage (LV) supply pad **130**. In embodiments, the second power-on-reset **304** monitors the PD1_LV SUPPLY at switch outputs (i) driven from low-voltage (LV) supply pad **130** when the first switch **308** is closed and the second switch **310** is open, (ii) driven from power supply pads **126** when the first switch **308** is open and the second switch **310** is closed, or (iii) driven from both low-voltage (LV) supply pad **130** and power supply pads **126** when both first switch **308** is closed and the second switch **310** is closed. In embodiments, the third power-on-reset **306** monitors the PD2_LV SUPPLY from power supply pads **126**.

(73) Power-on-reset circuit **300** allows the detection of the trip point of the power-on-reset of the third power domain **106** by monitoring the GPIO pad **318**. In embodiments, the first switch **308** is mapped to the low-power switch **118** in the system-on-chip **100**. In embodiments, the second switch **310** is mapped to the RUN mode switches **120**, **122**, and **124** in the system-on-chip **100**. It is noted that, although multiple RUN mode switches are shown in FIG. **1**, the number of switches is reduced to one in FIG. **3** for simplicity of the discussion.

(74) In embodiments, the first power-on-reset **302** is coupled to the low-voltage (LV) supply pad

130 of the first power domain **102** (PD0_LV Supply). The second power-on-reset **304** is coupled to the low-voltage (LV) supply at the switch outputs (i.e., between the first switch **308** and the second switch **310**) of the second power domain **104** (PD1_LV Supply). In embodiments, the third power-on-reset **306** is coupled to the low-voltage (LV) supply pads **126** of the third power domain **106** (PD2_LV Supply).

(75) In the power-on-reset circuit **300**, by selectively closing and opening the first switch **308** and the second switch **310**, each low-voltage (LV) supply from the three different power domains can be individually ramped down while keeping the other low-voltage (LV) supplies stable. Ramping down the individual low-voltage (LV) supply and monitoring the value at the GPIO pad **318**, allows the trip point detection for the power-on-reset for the third power domain **106**.

(76) In embodiments, to test the power-on-reset for the third power domain **106**, the first switch **308** (i.e., the low-power switch **118** in system-on-chip **100**) is configured to be in the closed position, and the second switch **310** (i.e., the RUN mode switches **120**, **122**, and **124** in system-on-chip **100**) is configured to be in the open position. Thus, the third power-on-reset **306** is coupled to the low-voltage (LV) supply pads **126** of the PD2_LV Supply.

(77) In this configuration, PD2_LV Supply is decoupled from PD0_LV Supply and PD1_LV Supply, while PD0_LV supply is coupled to the PD1_LV Supply—the PD0_LV Supply is provided from the low-voltage (LV) supply pad **130** for the first and second power domains.

(78) It is noted that the outputs of the first power-on-reset **302** and the second power-on-reset **304** are kept stable, with an output of “1”. Thus, the output of the AND gate **312** depends solely on the output of the third power-on-reset **306** which depends on the PD2_LV Supply voltage level provided from low-voltage supply pads **126**.

(79) Further, each level shifter **314** and **316** has a first input coupled to the PD2_LV Supply and a second input coupled to the high-voltage (HV) supply. During the power-on-reset test for the third power domain **106**, the value at the third input of the second level shifter **316** is forced to “0”.

(80) Initially, when the PD2_LV Supply is above the power-on-reset threshold of the third power domain **106**, the control signal (PWR_OK) at the output of the first level shifter **314** has a value equal to “1”. In response to the control signal (PWR_OK) being activated (i.e., equal to “1”), the GPIO pad **318** is operable and equal to the value forced at the third input of the second level shifter **316** or “0”.

(81) Next, the PD2_LV Supply is ramped down (i.e., by ramping down the voltages at power supply pad **126**), while PD0_LV Supply is kept stable. In response to the PD2_LV Supply dropping below the power-on-reset threshold of the third power domain **106**, the control signal (PWR_OK) at the output of the first level shifter **314** changes from “1” to “0”. As a result, the GPIO pad **318** is pulled up-transitioning from the value “0” to the value of “1”. Thus, by monitoring the GPIO pad **318** and detecting the transition from “0” to “1”, the trip point for the power-on-reset of the third power domain **106** is detected.

(82) FIG. **4** illustrates a block diagram of an embodiment power-on-reset circuit **400** for testing the power-on-reset trip-point of the second power domain **104** of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode. Power-on-reset circuit **400** includes a first power-on-reset **302**, a second power-on-reset **304**, a first switch **406**, a second switch **408**, a first level shifter **410**, a second level shifter **412**, and a GPIO pad **414**, which may (or may not) be arranged as shown. Power-on-reset circuit **400** may include additional components not shown.

(83) Power-on-reset circuit **400** allows the detection of the trip point of the power-on-reset of the second power domain **104** by monitoring the GPIO pad **414**. In embodiments, the first switch **406** is mapped to the low-power switch **118** in the system-on-chip **100**. In embodiments, the second switch **408** is mapped to the RUN mode switches **120**, **122**, and **124** in the system-on-chip **100**. It is noted that, although multiple RUN mode switches are shown in FIG. **1**, the number of switches is reduced to one in FIG. **4** for simplicity of the discussion.

(84) As the PD2_LV Supply is ramped down (the third power domain **106** is powered off), to test

the power-on-reset for the second power domain **104** (e.g., testing power domains that don't have a dedicated supply), and the first power domain **102** and the second power domain **104** are powered on. The input to the second power-on-reset **304** is initially coupled to the PD0_LV Supply by closing the first switch **406**. Further, each level shifter **410** and **412** has a first input coupled to the PD0_LV Supply and a second input coupled to the high-voltage (HV) supply.

(85) In this initial configuration, the control signal (PWR_OK) at the output of the first level shifter **410** has a value equal to "1". The value at the third input of the second level shifter **412** is equal to the output of the second power-on-reset **304**. The output of the second power-on-reset **304** depends on the PD1_LV supply at the first switch **406** output driven from the PD0_LV Supply voltage level provided from low-voltage (LV) supply pad **130** through the first switch **406**. In response to the control signal (PWR_OK) being activated (i.e., equal to "1"), the value at the GPIO pad **414** is equal to the value at the third input of the second level shifter **412** or "1".

(86) It is noted that the second switch **408** (e.g., RUN mode switches **120**, **122**, and **124**) is in the open position. Next, first switch **406** (e.g., low-power switch **118**) is configured to be in the open position and the value at the third input of the second level shifter **412** is equal to the PD1_LV Supply, which gets ramped down automatically as switches **406** and **408** remain open while the second power domain **104** is decoupled from the low-voltage (LV) supply pads of the first power domain **102** and the third power domain **106**. The control signal (PWR_OK) at the output of the first level shifter **410** remains at "1" as the PD0_LV Supply remains available and stable. In response to the PD1_LV Supply at the third input of the second level shifter **412** dropping below the power-on-reset threshold of the second power domain **104**, the logic value at the GPIO pad **414** transitions from "1" to "0". Thus, by monitoring the GPIO pad **414** and detecting the transition from "1" to "0", the trip point for the power-on-reset of the second power domain **104** is detected.

(87) FIG. **5** illustrates a block diagram of an embodiment power-on-reset circuit **500** for testing the power-on-reset trip-point of the first power domain **102** of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode. Power-on-reset circuit **500** includes a first power-on-reset **302**, a first switch **504**, a second switch **506**, a first level shifter **508**, a second level shifter **510**, and a GPIO pad **512**, which may (or may not) be arranged as shown. Power-on-reset circuit **500** may include additional components not shown.

(88) Power-on-reset circuit **500** allows the detection of the trip point of the power-on-reset of the first power domain **102** by monitoring the GPIO pad **512**. In embodiments, the first switch **504** is mapped to the low-power switch **118** in the system-on-chip **100**. In embodiments, the second switch **506** is mapped to the RUN mode switches **120**, **122**, and **124** in the system-on-chip **100**. It is noted that, although multiple RUN mode switches are shown in FIG. **1**, the number of switches is reduced to one in FIG. **5** for simplicity of the discussion.

(89) Initially, PD2_LV Supply and PD1_LV Supply are powered off. The input to the first power-on-reset **302** is coupled to the low-voltage (LV) supply pad **130** of the first power domain **102** (PD0_LV Supply) and isolated from other power supplies by opening switches **504** and **506**. Further, each level shifter **508** and **510** has a first input coupled to the PD0_LV Supply and a second input coupled to the high-voltage (HV) supply. During the power-on-reset test for the first power domain **102**, the value at the third input of the second level shifter **510** is forced to "0".

(90) In the initial configuration, as the PD0_LV Supply is above the power-on-reset threshold of the first power domain **102**, the control signal (PWR_OK) at the output of the first level shifter **508** has a value equal to "1". In response to the control signal (PWR_OK) being activated (i.e., equal to "1"), the value at the GPIO pad **512** is equal to the value at the third input of the second level shifter **510** or "0".

(91) Next, the PD0_LV Supply is ramped down. In response to the PD0_LV Supply dropping below the power-on-reset threshold of the first power domain **102**, the control signal (PWR_OK) at the output of the first level shifter **508** transitions from a value of "1" to "0". As a result, the GPIO pad **512** is pulled up—transitioning from the value "0" to the value of "1". Thus, by monitoring the

GPIO pad **512** and detecting the transition from "0" to "1", the trip point for the power-on-reset of the first power domain **102** is detected.

(92) FIG. **6** illustrates a block diagram of an embodiment circuit **600** for generating the isolation control signal (i.e., gating control signal) for the signals crossing from an unpowered domain to the power domain and used in, for example, circuits **300**, **400**, and **500**. Circuit **600** includes an AND gate **602** coupled to a level shifter **604**, which may (or may not) be arranged as shown. Additionally, circuit **600** may include additional components not shown.

(93) Circuit **600** is shown as a generic circuit where "x" represents the power domain. In embodiments, each power domain includes a dedicated circuit **600** to generate a respective isolation control signal (ISO_CTRL_LVx) (i.e., gating control signal) for signals entering its power domain from other power domains. The isolation control signal (i) allows signals from another power domain in response to the other power domain being powered and (ii) forces signals from another power domain to a safe state (i.e., logic off state—whether it is "0" or "1") in response to the other power domain being unpowered.

(94) In embodiments, in response to (i) the first power domain **102** being unpowered, (ii) the second power domain **104** being unpowered, (iii) the third power domain **106** being powered, and (iv) the isolation control signal (ISO_CTRL_LV2) being at a logic low, all signals from the second power domain **104** and the first power domain **102** entering the third power domain **106**, are forced to the logic off state (i.e., blocked) for that respective signal within the third power domain **106**. This configuration corresponds to, for example, the boot sequence in a dual power flow.

(95) In embodiments, in response to (i) the first power domain **102** being powered, (ii) the second power domain **104** being powered, (iii) the third power domain **106** being powered, and (iv) the isolation control signal (ISO_CTRL_LV2) being at a logic high, all signals from the second power domain **104** and the first power domain **102** entering the third power domain **106** are allowed to enter to the third power domain **106**.

(96) In embodiments, in response to (i) the first power domain **102** being unpowered, (ii) the third power domain **106** being unpowered, (iii) the second power domain **104** being powered, and (iv) the isolation control signal (ISO_CTRL_LV1) being at a logic low, all signals from the third power domain **106** and the first power domain **102** entering the second power domain **104**, are forced to the logic off state (i.e., blocked) for that respective signal within the second power domain **104**. This configuration corresponds to, for example, a low-power mode or the boot sequence in a dual power flow.

(97) In embodiments, in response to (i) the first power domain **102** being powered, (ii) the third power domain **106** being powered, (iii) the second power domain **104** being powered, and (iv) the isolation control signal (ISO_CTRL_LV1) being at a logic high, all signals from the third power domain **106** and the first power domain **102** entering the second power domain **104** are allowed to enter to the second power domain **104**.

(98) In embodiments, in response to (i) the second power domain **104** being unpowered, (ii) the third power domain **106** being unpowered, (iii) the first power domain **102** being powered, and (iv) the isolation control signal (ISO_CTRL_LV0) being at a logic low, all signals from the third power domain **106** and the second power domain **104** entering the first power domain **102**, are forced to the logic off state (i.e., blocked) for that respective signal within the first power domain **102**. This configuration corresponds to, for example, a standby mode.

(99) In embodiments, in response to (i) the first power domain **102** being powered, (ii) the third power domain **106** being powered, (iii) the second power domain **104** being powered, and (iv) the isolation control signal (ISO_CTRL_LV0) being at a logic high, all signals from the third power domain **106** and the second power domain **104** entering the first power domain **102** are allowed to enter to the first power domain **102**.

(100) In embodiments, the first input to the AND gate **602** is the first voltage level (POR_LVx) of the low-voltage (LV) supply in a first configuration, where the HV supply becomes available

before low-voltage (LV) supplies, and where the HV supply provides for the LV supplies.
(101) In embodiments, the second input to the AND gate **602** is the second voltage level (POR_Vddreg_byLV) of the low-voltage (LV) supply in a second configuration, where the HV supply becomes available after the low-voltage (LV) supplies, the HV supply does not provide the LV supplies, and the voltage for the LV supplies are provided externally.
(102) Level shifter **604** is a high-voltage (HV) to low-voltage (LV) level shifter. The first transistor input of the level shifter **604** is coupled to the high-voltage (HV) supply.
(103) In an embodiment, to generate the isolation control signal (ISO_CTRL_LV0) (i.e., isolation control of the signals entering the first power domain **102**) to the first power domain **102**, the second transistor input of the level shifter **604** is coupled to the first power-on-reset **302** output.
(104) In another embodiment, to generate the isolation control signal (ISO_CTRL_LV1) (i.e., isolation control of the signals entering the second power domain **104**) to the second power domain **104**, the second transistor input of the level shifter **604** is coupled to the second power-on-reset **304** output.
(105) In yet another embodiment, to generate the isolation control signal (ISO_CTRL_LV2) (i.e., isolation control of the signals entering third power domain **106**) to the third power domain **106**, the second transistor input of the level shifter **604** is coupled to the third power-on-reset **306** output.
(106) Thus, depending on the power-on-reset and the PDx_LV_SUPPLY, the level shifter **604** level shifts the ISO_CTRL_HVx (i.e., from the HV supply level of the first power domain **102**) to ISO_CTRL_LVx (i.e., to the associated LV supply level of that particular power domain). The level-shifted isolation control signal (ISO_CTRL_LVx) is used as the isolation control (i.e., gating control) of the signals entering that particular power domain from other power domains.
(107) In embodiments, the level-shifted isolation control signal (ISO_CNTRL_LVx) is forced to a logic high “1” when the output of the AND gate **602** is at a logic high “1”. Thus, the associated LV supply level of that particular power domain is set above the corresponding power-on-reset threshold.
(108) In embodiments, the level-shifted isolation control signal (ISO_CONTRL_LVx) is forced to a logic low “0” when the output of the AND gate **602** is at a logic low “0”. Thus, the associated LV supply level of that particular power domain is set below the corresponding power-on-reset threshold.
(109) FIG. **7** illustrates a block diagram of an embodiment circuit **700** used to illustrate how the input value to the third input of the second level shifter **510** is forced to the value of “0”. Circuit **700** includes the power-on-reset **302**, a test data register (TDR) **702**, the first level shifter **508**, an inverter **704**, the second level shifter **510**, and the GPIO pad **512**, which may (or may not) be arranged as shown. Circuit **700** may include additional components not shown.
(110) In embodiments, during the power-on-reset test mode, the test data register **702** is loaded with a value of “1”. Therefore, the output of inverter **704** is set to the value of “0”. Thus, the third input to the second level shifter **510** is forced to the value of “0”. It is noted that the test data register **702** is reset to “0” when the power-on-reset test is not being performed. In embodiments, the test data register **702** is reset or loaded with the value of “1” using a JTAG.
(111) FIG. **8** illustrates a block diagram of an embodiment circuit **800** used to illustrate the input value to the third input of the second level shifter **316**. Circuit **800** includes a first AND gate **306** (i.e., the third power-on-reset **306**), an HV to LV level shifter **804**, a test data register (TDR) **806**, a second AND gate **808**, an inverter **810**, a third AND gate **812**, the first level shifter **314**, the second level shifter **316**, and the GPIO pad **318**, which may (or may not) be arranged as shown. Further, circuit **800** may include additional components not shown.
(112) First AND gate **306** (third power-on-reset) and the HV to LV level shifter **804** are arranged as circuit **600**. The output signal (ISO_CTRL_LV2) corresponds to the value ISO_CTRL_HV2 and the value at the output of first AND gate **306** (the third power-on-reset **306** output).
(113) In embodiments, the input to the first level shifter **314** is the output of the second AND gate

808. The output of the second AND gate **808** corresponds to the values of the first power-on-reset **302** output, the value at ISO_CTRL_HV2, and the value at the third power-on-reset **306** output.
(114) As the value at first power-on-reset **302** is high during the testing of the power-on-reset of the third power domain **106**, also the ISO_CTRL_HV2 is at a logic high "1" (i.e., as the HV supply is stable), the output of the AND gate **808** corresponds to the value of isolation control signal (ISO_CTRL_LV2) from the HV to LV level shifter **804**. The value of isolation control signal (ISO_CTRL_LV2) is an AND logic of low-voltage power-on-resets monitoring the supplies (PD2_LV SUPPLY) from low-voltage (LV) supply pads **126**, as detailed in FIG. **6** for the third power domain **106**.
(115) The input to the second level shifter **316** is the AND logic of the isolation control signal (ISO_CTRL_LV2) and the inverted value loaded at the test data register **806**. During the testing of the power-on-reset of the third power domain **106**, the test data register **806** is loaded with a value of "1". Thus, the value of the third input to the second level shifter **316** is forced to "0".
(116) FIG. **9** illustrates a flow chart of an embodiment method **900** to test power-on-resets of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode.
(117) At step **902**, the system-on-chip is fully powered up, low-power switch **118** and RUN mode switches **120**, **122**, and **124** are closed. The test data register **702** and **806** are reset to "0". The output of the control signals (PWR_OK) for the various power domains have a value equal to "1" when both the HV and LV supplies are available.
(118) At step **904**, the testing of the power-on-reset of the third power domain **106** begins. The test data register **806** is loaded with the value of "1". The third power domain **106** is decoupled from the first power domain **102** and the second power domain **104** by opening the RUN mode switches **120**, **122**, and **124**.
(119) At step **906**, the low-voltage (LV) supply of the third power domain **106** (PD2_LV Supply) is ramped down and the GPIO pad **318** is monitored. Before the PD2_LV Supply is ramped down, the value at the GPIO pad **318** is "0".
(120) At step **908**, while PD2_LV Supply is ramping down, in response to detecting a transition from "0" to "1" at the GPIO pad **318**, the trip point for the power-on-reset of the third power domain **106** is detected.
(121) At step **910**, the second power domain **104** is decoupled from the first power domain **102** by opening the low-power switch **118**.
(122) At step **912**, the low-voltage (LV) supply of the second power domain **104** (PD1_LV Supply) is ramped down and the GPIO pad **414** is monitored. Before the PD1_LV Supply is ramped down, the value at the GPIO pad **414** is "1".
(123) At step **914**, while PD1_LV Supply is ramping down, in response to detecting a transition from "1" to "0" at the GPIO pad **414**, the trip point for the power-on-reset of the second power domain **104** is detected.
(124) At step **916**, the low-voltage (LV) supply of the first power domain **102** (PD0_LV Supply) is ramped down and the GPIO pad **512** is monitored. Before the PD0_LV Supply is ramped down, the value at the GPIO pad **512** is "0".
(125) At step **918**, while PD0_LV Supply is ramping down, in response to detecting a transition from "0" to "1" at the GPIO pad **512**, the trip point for the power-on-reset of the first power domain **102** is detected.
(126) FIG. **10** illustrates a block diagram of an embodiment processing system **1000**. As shown, the processing system **1000** includes a processor **1002**, a memory **1004**, and an interface **1006**, which may (or may not) be arranged as shown. In embodiments, the processing system **1000** may be coupled to the system-on-chip **100** for testing the various power-on-resets of the various power domains. In embodiments, controller **108** corresponds to the processing system **1000**.
(127) The processing system **1000** may include additional components not depicted, such as long-term storage (e.g., non-volatile memory, etc.), measurement devices, and the like.

(128) Processor **1002** may be any component or collection of components adapted to perform computations or other processing related tasks, as disclosed herein. Memory **1004** may **1004** may be any component or collection of components adapted to store programming or instructions for execution by the processor **1002**. In an embodiment, memory **1004** includes a non-transitory computer-readable medium.

(129) Interface **1006** may be any component or collection of components that allow the processor **1002** to communicate with other devices/components or a user. For example, interface **1006** may be adapted to allow a user or device (e.g., personal computer (PC), etc.) to interact/communicate with the processing system **1000**.

(130) FIG. **11** illustrates a block diagram of an embodiment circuit **1100** for testing power-on-resets of a system-on-chip with a multi-power domain architecture operating under a dual power flow mode. Circuit **1100** includes a first level shifter **1102**, a second level shifter **1104**, a third level shifter **1106**, a multiplexer **1108**, a first pad **1110**, a fourth level shifter **1112**, a second pad **1114**, a fifth level shifter **1116**, a sixth level shifter **1118**, a third pad **1120**, and a switch **1122**, which may (or may not) be arranged as shown. Further, circuit **1100** may include additional components not shown. A probe point within the die is shown as element **1124** at the output of the third pad **1120**.

(131) The inputs of the first level shifter **1102**, the second level shifter **1104**, and the third level shifter **1106** are, respectively, coupled to a representative logic value of the low-voltage supply of the first power domain (POR_LV0), the low-voltage supply of the second power domain (POR_LV1), and the low-voltage supply of the third power domain (POR_LV2).

(132) The outputs of the first level shifter **1102**, the second level shifter **1104**, and the third level shifter **1106** are coupled to the multiplexer **1108**.

(133) The first pad **1110** and the second pad **1114** are pads in the first power domain. Based on the values at the first pad **1110** and the second pad **1114** (configured in test mode to operate as power-on-reset test control pads), the multiplexer **1108** selects the output of one of the level shifters.

(134) Initially, the system-on-chip is fully powered up and the low-power switch **118** and the RUN mode switches **120**, **122**, and **124** are in a closed position. A test data register coupled to the input of the third pad **1120** through an inverter is reset at “0”—the value at the third pad **1120** is “1”. The outputs of the first pad **1110** and the second pad **1114** are set to “0”.

(135) During the power-on-reset test mode, the test data register is loaded with a value of “1” using, for example, a JTAG. To test the power-on-reset for the low-voltage supply of the third power domain **106**, the first pad **1110** and the second pad **111** are forced to “01”, which results in the multiplexer **1108** selecting the low-voltage power-on-reset monitoring the supply of the third power domain **106** at its output. The third power domain **106** is decoupled from the first power domain **102** and the second power domain **104** by opening the RUN mode switches **120**, **122**, and **124**. The low-voltage supply of the third power domain **106** is ramped down. In response to the low-voltage supply of the third power domain **106** reaching the power-on-reset threshold of the third power domain **106**, the third pad **1120** transitions from “1” to “0”. The transition point is the power-on-reset trip point of the low-voltage supply of the third power domain **106**.

(136) Next, to test the power-on-reset for the low-voltage supply of the second power domain **104**, the first pad **1110** and the second pad **1114** are forced to “10”, which results in the multiplexer **1108** selecting the low-voltage power-on-reset monitoring the supply of the second power domain **104** at its output. The second power domain **104** is decoupled from the first power domain **102** by opening the low-power switch **118**. The low-voltage supply of the second power domain **104** is ramped down. In response to the low-voltage supply of the second power domain **104** reaching the power-on-reset threshold of the second power domain **104**, the third pad **1120** transitions from “1” to “0”. The transition point is the power-on-reset trip point of the low-voltage supply of the second power domain **104**.

(137) Finally, to test the power-on-reset for the low-voltage supply of the first power domain **102**, the first pad **1110** and the second pad **1114** are forced to “11”, which results in the multiplexer **1108**

selecting the low-voltage power-on-reset monitoring the supply of the first power domain **102** at its output. The low-voltage supply of the first power domain **102** is ramped down. In response to the low-voltage supply of the first power domain **102** reaching the power-on-reset threshold of the first power domain **102**, the third pad **1120** transitions from “0” to “1”. The transition point is the power-on-reset trip point of the low-voltage supply of the first power domain **102**.
(138) A similar circuit to circuit **1100** can be used to test other power-on-resets of the various power domains. For example, in embodiments, the inputs of the first level shifter **1102**, the second level shifter **1104**, and the third level shifter **1106** are, respectively, coupled to a representative logic value of the low-voltage supply of the first power domain (POR_Vddreg_byLV0), the low-voltage supply of the second power domain (POR_Vddreg_byLV0), and the low-voltage supply of the third power domain (POR_Vddreg_byLV0), as discussed in the discussion for the circuit **600**.
(139) It is noted that all steps outlined are not necessarily required and can be optional. Further, changes to the arrangement of the steps, removal of one or more steps and path connections, and addition of steps and path connections are similarly contemplated.
(140) Although the description has been described in detail, it should be understood that various changes, substitutions, and alterations may be made without departing from the spirit and scope of this disclosure as defined by the appended claims. The same elements are designated with the same reference numbers in the various figures. Moreover, the scope of the disclosure is not intended to be limited to the particular embodiments described herein, as one of ordinary skill in the art will readily appreciate from this disclosure that processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, may perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.
(141) The specification and drawings are, accordingly, to be regarded simply as an illustration of the disclosure as defined by the appended claims, and are contemplated to cover any and all modifications, variations, combinations, or equivalents that fall within the scope of the present disclosure.

Claims

1. A method for testing multiple power-on-resets in a system-on-chip with a multi-power domain architecture operating under a dual power flow mode, the method comprising: powering up the system-on-chip to full power mode; decoupling a third power domain from a first power domain and a second power domain; monitoring a general purpose input/output (GPIO) pad of the third power domain during a ramping down of a supply of the third power domain; and detecting a logic transition at the GPIO pad of the third power domain corresponding to a trip-point of the power-on-reset of the third power domain.
2. The method of claim 1, further comprising: decoupling the second power domain from the first power domain; monitoring a GPIO pad of the first power domain during a ramping down of a supply of the second power domain; and detecting a logic transition at the GPIO pad of the first power domain corresponding to a power-on-reset trip-point of the second power domain.
3. The method of claim 2, further comprising: monitoring a second GPIO pad of the first power domain during a ramping down of a supply of the first power domain; and detecting a logic transition at the second GPIO pad of the first power domain corresponding to a trip-point of the power-on-reset of the first power domain.
4. The method of claim 1, wherein a supply of the second power domain and a supply of the first power domain remain stable during the ramping down of the supply of the third power domain.
5. The method of claim 1, wherein the system-on-chip with the multi-power domain architecture

operating under dual power flow mode has a first power flow direction during a power up or a reset sequence different from a second power flow direction during an exiting of a standby mode.
6. The method of claim 5, wherein the first power flow direction is a sequential powering up of the third power domain, the second power domain, and the first power domain.
7. The method of claim 6, wherein the second power flow direction is a sequential powering up of the second power domain and the third power domain, wherein the first power domain remains on during the standby mode.
8. The method of claim 1, wherein the third power domain is used to perform full-power operations of the system-on-chip, wherein the second power domain is used to perform low-power operations of the system-on-chip, and wherein the first power domain is used to enter the system-on-chip into a standby mode.
9. A system-on-chip, comprising: a first power domain for operating the system-on-chip in a standby mode, the first power domain comprising a first switch; a second power domain for operating the system-on-chip in a low-power mode, the first switch configured to couple the first power domain to the second power domain, the second power domain comprising a second switch; a third power domain for operating the system-on-chip in a full operation mode, the second switch configured to couple the third power domain to the second power domain and the first switch configured to couple the second power domain to the first power domain, the third power domain comprising: a first level shifter configured to: generate a control signal with a logic level high in response to a supply of the third power domain being greater than a first threshold, the system-on-chip being fully powered up, and the second switch being in an open position, and generate the control signal with a logic level low in response to the supply of the third power domain being less than the first threshold, the system-on-chip being powered up in the first power domain and the second power domain and unpowered in the third power domain, and the second switch being in the open position; and a second level shifter configured to: receive the control signal, generate an output signal with a logic level low in response to the control signal having a logic level high, and generate the output signal with a logic level high in response to the control signal having a logic level low.
10. The system-on-chip of claim 9, further comprising a GPIO pad coupled to an output of the second level shifter, wherein by monitoring a transition from the logic level low to a logic level high at the GPIO pad during a ramping down of the supply of the third power domain, a power-on-reset trip point of the supply of the third power domain is detected.
11. The system-on-chip of claim 9, wherein the second power domain comprises a third level shifter configured to: generate a second output signal with a logic level high in response to a supply of the second power domain being greater than a second threshold, and generate the second output signal with a logic level low in response to the supply of the second power domain being less than the second threshold.
12. The system-on-chip of claim 11, further comprising a GPIO pad coupled to an output of the third level shifter, wherein by monitoring a transition from the logic level high to a logic level low at the GPIO pad during a ramping down of the supply of the second power domain, a power-on-reset trip point of the supply of the second power domain is detected.
13. The system-on-chip of claim 11, wherein the first power domain comprises: a fourth level shifter configured to: generate a second control signal with a logic level high in response to a power-on-reset of the first power domain being greater than a third threshold, and generate the second control signal with a logic level low in response to the power-on-reset of the first power domain being less than the third threshold; and a fifth level shifter configured to: receive the second control signal, generate a third output signal with a logic level low in response to the second control signal having a logic level high, and generate the third output signal with a logic level high in response to the second control signal having a logic level low.
14. The system-on-chip of claim 13, further comprising a GPIO pad coupled to an output of the

fifth level shifter, wherein by monitoring a transition from the logic level low to a logic level high at the GPIO pad during a ramping down of the supply of the first power domain, a power-on-reset trip point of the supply of the first power domain is detected.

15. A method for testing multiple power-on-resets in a system-on-chip with a multi-power domain architecture operating under a dual power flow mode, the method comprising: having a first power domain for operating the system-on-chip in a standby mode, the first power domain comprising a first switch; having a second power domain for operating the system-on-chip in a low-power mode, the first power domain couplable to the second power domain using the first switch, and the second power domain comprising a second switch; a third power domain for operating the system-on-chip in a full operation mode, the third power domain couplable to the second power domain using the second switch and further couplable to the first power domain using the first switch; operating the system-on-chip in a power-on-reset test mode; opening the second switch to decouple the third power domain from the first power domain and the second power domain during full power up of the system-on-chip; loading a test data register with a bit having a logic level high; ramping down of a supply of the third power domain; and detecting a logic level transition at an output of a first level shifter, the first level shifter coupled to an inverted value of the test data register and controlled by a control signal generated by a second level shifter, the second level shifter generating a first control signal in response to the supply of the third power domain being greater than a threshold and generating a second control signal in response to the supply of the third power domain being less than the threshold.

16. The method of claim 15, wherein the detecting the logic level transition at the output of the first level shifter corresponds to a power-on-reset trip point of the supply of the third power domain.

17. The method of claim 15, further comprising: opening the first switch to decouple the second power domain from the first power domain; ramping down of a supply of the second power domain; and detecting a logic level transition at an output of a third level shifter, the third level shifter generating a first output signal with a logic level high in response to the supply of the second power domain being greater than a second threshold and generating a second output signal with a logic level low in response to the supply of the second power domain being less than the second threshold.

18. The method of claim 17, wherein the detecting the logic level transition at the output of the third level shifter corresponds to a power-on-reset trip point of the supply of the second power domain.

19. The method of claim 17, further comprising: loading a second test data register with a bit having a logic level high; ramping down of a supply of the first power domain; and detecting a logic level transition at an output of a fourth level shifter, the fourth level shifter coupled to an inverted value of the second test data register and controlled by a third control signal generated by a fifth level shifter, the fifth level shifter generating a fourth control signal in response to the supply of the first power domain being greater than a third threshold and generating a fifth control signal in response to the supply of the first power domain being less than the third threshold.

20. The method of claim 19, wherein the detecting the logic level transition at the output of the fourth level shifter corresponds to a power-on-reset trip point of the supply of the first power domain.